UNIT 1 · 8 hours

Electrical Systems and Analogous Systems



Introduction and Mathematical Modelling of Physical Systems

Control Systems (22EEG53)
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Learning objectives

- Derive transfer functions of RLC networks
- Write the force–voltage and force–current analogies
- Convert a mechanical system into an analogous electrical circuit
- Use the analogy to transfer design insight

Syllabus context

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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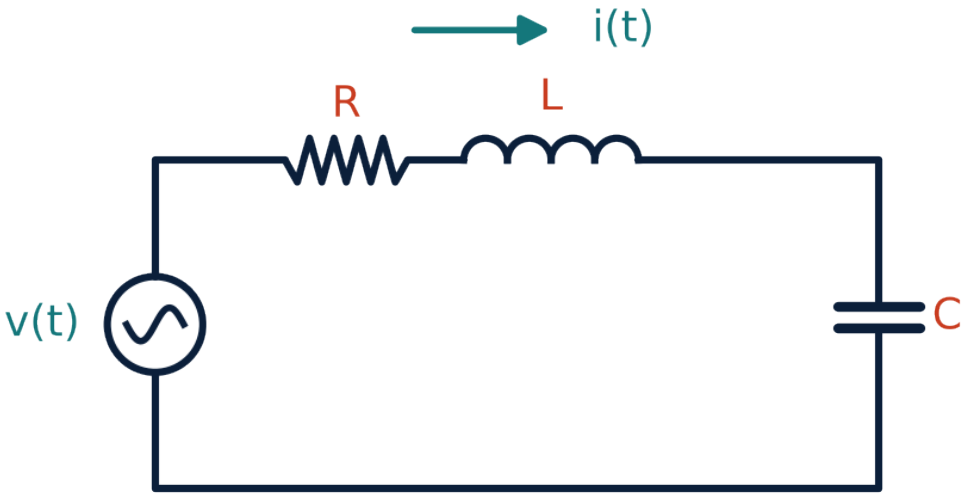
Electrical elements

Element	v–i relation	Laplace
Resistor R	$v = R\,i$	$R\,I(s)$
Inductor L	$v = L\,di/dt$	$L\,s\,I(s)$
Capacitor C	$v = (1/C)\int i\,dt$	$I(s)/(C\,s)$

Kirchhoff's laws (KVL, KCL) give the differential equations of a network.

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Series RLC network



- KVL: $v = R\,i + L\,di/dt + (1/C)\int i\,dt$.
- Output across C: $v_c = (1/C)\int i\,dt$.
- Laplace: $V(s) = (R + Ls + 1/Cs)\,I(s)$.
- $V_c(s)/V(s) = 1/(LC\,s^2 + RC\,s + 1)$.

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RLC parameters

$$\frac{V_c}{V} = \frac{1}{LC\,s^2 + RC\,s + 1}$$

transfer function

$$\omega_n = \frac{1}{\sqrt{LC}}$$

natural frequency

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

damping ratio

- Compare with the mechanical system: M ↔ L, B ↔ R, K ↔ 1/C.

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Force–voltage analogy (F–V)

Mechanical (translation)	Electrical	Equation
Force f	Voltage e	f ↔ e
Mass M	Inductance L	M ↔ L
Damper B	Resistance R	B ↔ R
Spring K	Reciprocal capacitance 1/C	K ↔ 1/C
Displacement x	Charge q	x ↔ q
Velocity v	Current i	v ↔ i

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Force–current analogy (F–I)

Mechanical (translation)	Electrical	Equation
Force f	Current i	f ↔ i
Mass M	Capacitance C	M ↔ C
Damper B	Conductance 1/R	B ↔ 1/R
Spring K	Reciprocal inductance 1/L	K ↔ 1/L
Displacement x	Flux linkage ψ	x ↔ ψ
Velocity v	Node voltage e	v ↔ e

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Numerical: analogous circuits

Problem A system has M = 2 kg, B = 4 N·s/m, K = 8 N/m. Find the F–V and F–I analogous circuit values.

- F–V (series RLC): L = M = 2 H; R = B = 4 Ω; C = 1/K = 0.125 F
- F–I (parallel RLC): C = M = 2 F; 1/R = B → R = 0.25 Ω; 1/L = K → L = 0.125 H
- Both give ωn = √(K/M) = 2 rad/s

Answer: F–V: L = 2 H, R = 4 Ω, C = 0.125 F. F–I: C = 2 F, R = 0.25 Ω, L = 0.125 H

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Key takeaways

- RLC: $V_c/V = 1/(LCs^2 + RCs + 1)$, $\omega_n = 1/\sqrt{(LC)}$
- F–V analogy: M ↔ L, B ↔ R, K ↔ 1/C, f ↔ e
- F–I analogy: M ↔ C, B ↔ 1/R, K ↔ 1/L, f ↔ i
- Analogies let us study mechanical systems with circuits

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Practice questions

Q1. Derive Vc/V for a series RLC network.

Q2. Draw the F–V analogue of a mass-spring-damper system.

Q3. M = 5, B = 10, K = 20: find F–I analogue parameters.

Q4. State the table of analogous quantities.